

Double Integrals



Iterated integrals over rectangles

- 1 Evaluate $\int_0^2 \int_0^3 (x^2 + y) dy dx$.
 - 2 Evaluate $\int_1^2 \int_0^1 6xy^2 dx dy$.
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Double integrals over general regions

- 3 Evaluate $\int_0^1 \int_0^x (x + y) dy dx$, integrating over the triangular region $0 \leq y \leq x \leq 1$.
 - 4 Use a double integral to find the volume under $f(x, y) = 4 - x - y$ over the rectangle $0 \leq x \leq 1, 0 \leq y \leq 1$.
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Reversing the order of integration

- 5 Evaluate $\int_0^1 \int_y^1 e^{x^2} dx dy$ by first reversing the order of integration (the region is $0 \leq y \leq x \leq 1$).
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Double integrals in polar coordinates

- 6 Evaluate $\iint_D (x^2 + y^2) dA$ where D is the disk $x^2 + y^2 \leq 4$, by converting to polar coordinates ($dA = r dr d\theta$).
 - 7 Find the area of the region enclosed by $r = 2\cos \theta$ using a double integral in polar coordinates.
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Average value over a region

- 8 Find the average value of $f(x, y) = xy$ over the rectangle $0 \leq x \leq 2, 0 \leq y \leq 3$ (average value $= \frac{1}{A} \iint_D f dA$, where A is the region's area).
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Mass and center of a region

- 9 A lamina occupies the rectangle $0 \leq x \leq 2, 0 \leq y \leq 1$ with density $\rho(x, y) = x + y$. Find its total mass, $M = \iint_D \rho dA$.

- 10 For a lamina occupying $0 \leq x \leq 1$, $0 \leq y \leq 1$ with constant density $\rho = 1$, find the x -coordinate of the center of mass, $\bar{x} = \frac{1}{M} \iint_D x \, dA$ (where M is the total mass).
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Triple integral setup (a first look)

- 11 Set up (do not evaluate) the triple integral for the volume of the solid bounded by $0 \leq x \leq 1$, $0 \leq y \leq 2$, $0 \leq z \leq 3$.
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Double integrals with a variable upper limit

- 12 Evaluate $\int_0^1 \int_0^{y^2} 3x \, dx \, dy$, integrating over the region $0 \leq x \leq y^2$, $0 \leq y \leq 1$.
- 13 Set up and evaluate the double integral for the volume under $f(x, y) = xy^2$ over the triangular region bounded by $y = x$, $y = 0$, and $x = 2$.
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An applied double-integral problem

- 14 A metal plate occupies the region $0 \leq x \leq 3$, $0 \leq y \leq 2$, with temperature distribution $T(x, y) = 50 - x^2 - y^2$ (degrees Celsius). Find the average temperature over the plate.
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Volume between two surfaces

- 15 Find the volume of the solid between $z = 8 - x^2 - y^2$ and $z = 0$ over the region where $z \geq 0$ requires $x^2 + y^2 \leq 8$, using polar coordinates.
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Double integrals with exponential integrands

- 16 Evaluate $\int_0^1 \int_0^1 x e^{xy} \, dy \, dx$ by integrating with respect to y first.
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A capstone applied double-integral problem

- 17 An architect is designing a swimming pool whose floor depth varies according to $d(x, y) = 1 + 0.2x + 0.1y$ (meters) over a rectangular footprint $0 \leq x \leq 10$, $0 \leq y \leq 5$ (meters). Set up and evaluate the double integral for the total volume of water needed to fill the pool, then find the average depth by dividing by the pool's surface area.

- 18 For the double integral $\iint_D (2x + y) \, dA$ where D is the region bounded by $y = x^2$ and $y = x$, complete each step:
- a Find the intersection points of the two curves to determine the bounds.
 - b Set up the iterated integral with x as the outer variable.
 - c Evaluate the integral.